

Domácí úkol 4.2

Wednesday 25 February 2026 17:54

$$\begin{aligned} p(0) &= 1 \Leftrightarrow 1 = a_0 \\ p(1) &= 2 \Leftrightarrow 2 = a_4 + a_3 + a_2 + a_1 + a_0 \\ p(2) &= -3 \Leftrightarrow -3 = 16a_4 + 8a_3 + 4a_2 + 2a_1 + a_0 \end{aligned} \rightarrow \begin{pmatrix} a_4 & a_3 & a_2 & a_1 & a_0 \\ 1 & 1 & 1 & 1 & 1 \\ 16 & 8 & 4 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$$

$$a_{\text{ch}} = A^T z, \text{ kde } z \text{ je řešení soustavy } AA^T z = b \rightarrow \begin{pmatrix} 1 & 1 & 1 & 1 \\ 16 & 8 & 4 & 2 \end{pmatrix} \begin{pmatrix} 1 & 16 \\ 1 & 8 \\ 1 & 4 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$$

$$\begin{aligned} \frac{1}{2} \begin{pmatrix} 4 & 30 & 1 \\ 30 & 340 & -4 \end{pmatrix} &\sim \begin{pmatrix} 2 & 15 & 1/2 \\ 30 & 340 & -4 \end{pmatrix} \sim \begin{pmatrix} 2 & 15 & 1/2 \\ 0 & 115 & -115 \end{pmatrix} \rightarrow \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1/10 \end{pmatrix} \\ &\quad \downarrow \quad \downarrow \\ &\quad 1 \quad -1/10 \end{aligned}$$

$$a_{\text{ch}} = \begin{pmatrix} 1 & 16 \\ 1 & 8 \\ 1 & 4 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -1/10 \end{pmatrix} = \begin{pmatrix} -6/10 \\ 2/10 \\ 6/10 \\ 8/10 \end{pmatrix} = \begin{pmatrix} -3/5 \\ 1/5 \\ 3/5 \\ 4/5 \end{pmatrix} \rightarrow \text{tedy polynom splňující zadání je: } p(x) = -\frac{3}{5}x^4 + \frac{1}{5}x^3 + \frac{3}{5}x^2 + \frac{4}{5}x + 1$$